

The Ellenberg–Gijswijt cap-set bound via slice rank

1 Preliminaries

Throughout, $\mathbb{F}_3 = \{0, 1, 2\}$ denotes the field with three elements and $n \geq 1$ an integer. For a vector of exponents $a = (a_1, \dots, a_n) \in \{0, 1, 2\}^n$ we write $x^a := \prod_{i=1}^n x_i^{a_i}$ for the corresponding monomial on $\mathbb{F}_3^n$, of total degree $\deg(x^a) = \sum_{i=1}^n a_i$. Because every $u \in \mathbb{F}_3$ satisfies $u^3 = u$, each polynomial function $\mathbb{F}_3^n \rightarrow \mathbb{F}_3$ is represented by a unique $\mathbb{F}_3$ -linear combination of such *reduced* monomials (each individual exponent in $\{0, 1, 2\}$); we identify functions with their reduced representatives. Throughout, $c, C, C_1, \dots$ denote universal positive constants whose value may change from line to line and depend on no parameter.

Definition 1.1 (Low-degree monomial space and its dimension). For a real threshold $\theta \geq 0$, let

$$V_{\leq \theta} := \text{span}_{\mathbb{F}_3} \{x^a : a \in \{0, 1, 2\}^n, \sum_{i=1}^n a_i \leq \theta\},$$

the space of reduced polynomial functions of total degree at most θ , and let

$$M_n := \dim_{\mathbb{F}_3} V_{\leq 2n/3} = \#\{a \in \{0, 1, 2\}^n : \sum_{i=1}^n a_i \leq 2n/3\}.$$

Definition 1.2 (Cap set). A set $A \subseteq \mathbb{F}_3^n$ is a *cap set* (is *three-term-progression free*) if the only solutions of $x + y + z = 0$ with $x, y, z \in A$ are the diagonal solutions $x = y = z$.

Remark 1.3 (Equivalence with the geometric definition). The usual definition asks that A contain no three distinct collinear points, i.e. no nontrivial three-term arithmetic progression $\{x, x+r, x+2r\}$ with $r \neq 0$. Over $\mathbb{F}_3$ a three-term progression x, y, z (with $y - x = z - y$) satisfies $x + z = 2y = -y$, hence $x + y + z = 0$; conversely $x + y + z = 0$ forces $y - x = z - y$. Thus x, y, z form a progression iff $x + y + z = 0$, and the progression is nontrivial iff the three points are not all equal. Definition 1.2 is therefore equivalent to the geometric definition.

Assumption 1.4 (Cap set in $\mathbb{F}_3^n$). $A \subseteq \mathbb{F}_3^n$ is a cap set in the sense of Definition 1.2: for all $x, y, z \in A$, $x + y + z = 0 \implies x = y = z$.

Remark 1.5 (Mildness). Assumption 1.4 is the defining property of the object under study and imposes no regularity beyond three-term-progression freeness. No bound on $|A|$, no structural hypothesis, and no constraint on n is assumed; the conclusion $|A| \leq 3M_n$ holds for every cap set.

2 Slice rank and its basic properties

Let $\mathbb{F}$ be a field and $X_1, \dots, X_k$ finite sets. A function $T : \prod_{j=1}^k X_j \rightarrow \mathbb{F}$ is called a *slice* if there is a coordinate $i \in [k]$, a function $f : X_i \rightarrow \mathbb{F}$, and a function $g : \prod_{j \neq i} X_j \rightarrow \mathbb{F}$ such that, writing $\widehat{x}_i$ for the tuple with the i -th entry omitted,

$$T(x_1, \dots, x_k) = f(x_i) g(\widehat{x}_i) \quad \text{for all } (x_1, \dots, x_k). \quad (1)$$

The *slice rank* $\text{sr}(T)$ is the least integer $r \geq 0$ such that T is a sum of r slices (and $\text{sr}(0) = 0$). For $k = 2$, sr is the ordinary rank of the matrix $(T(x_1, x_2))$.

Lemma 2.1 (Slice rank: subadditivity, diagonal lower bound, restriction). *Let $\mathbb{F}$ be a field and $X, X_1, \dots, X_k$ finite sets. The slice rank satisfies the following.*

- (a) (Subadditivity.) *For all $T, T' : \prod_j X_j \rightarrow \mathbb{F}$ and all $\lambda \in \mathbb{F} \setminus \{0\}$, $\text{sr}(T + T') \leq \text{sr}(T) + \text{sr}(T')$ and $\text{sr}(\lambda T) = \text{sr}(T)$.*
- (b) (Diagonal lower bound.) *Let $D : X^k \rightarrow \mathbb{F}$ be the diagonal tensor $D(x_1, \dots, x_k) = \sum_{a \in X} c_a \mathbf{1}[x_1 = \dots = x_k = a]$ with $c_a \neq 0$ for every $a \in X$. Then $\text{sr}(D) = |X|$.*
- (c) (Restriction monotonicity.) *For $A_j \subseteq X_j$ and $T : \prod_j X_j \rightarrow \mathbb{F}$, the restriction $T|_{\prod_j A_j}$ satisfies $\text{sr}(T|_{\prod_j A_j}) \leq \text{sr}(T)$.*

Proof. We prove the three parts in turn.

Part 2.1a (subadditivity). If $T = \sum_{i=1}^r S_i$ and $T' = \sum_{i=1}^{r'} S'_i$ are slice decompositions with $r = \text{sr}(T)$ and $r' = \text{sr}(T')$, then $T + T' = \sum_i S_i + \sum_i S'_i$ is a sum of $r + r'$ slices, so $\text{sr}(T + T') \leq r + r'$. Scaling a slice Eq. (1) by $\lambda \neq 0$ gives the slice $(\lambda f)(x_i) g(\hat{x}_i)$, so $\text{sr}(\lambda T) \leq \text{sr}(T)$; applying this to λ^{-1} gives the reverse inequality, hence equality.

Part 2.1b (diagonal lower bound). The upper bound $\text{sr}(D) \leq |X|$ is the decomposition

$$D(x_1, \dots, x_k) = \sum_{a \in X} c_a \mathbf{1}[x_1 = a] \cdot \mathbf{1}[x_2 = \dots = x_k = a],$$

a sum of $|X|$ slices. For the lower bound $\text{sr}(D) \geq |X|$ we argue by induction on k , proving the stronger claim: *every diagonal tensor $D'(x_1, \dots, x_k) = \sum_{a \in X'} c_a \mathbf{1}[x_1 = \dots = x_k = a]$ with $X' \subseteq X$ and all $c_a \neq 0$ has $\text{sr}(D') \geq |X'|$.* For the base case $k = 2$, sr is matrix rank and D' is a diagonal matrix with $|X'|$ nonzero entries, of rank exactly $|X'|$. Assume the claim for $k - 1$ and let $D' = \sum_{i=1}^r f_i(x_{c_i}) g_i(\hat{x}_{c_i})$ be any slice decomposition. Let $S \subseteq [r]$ index the slices whose special coordinate is the last one, $c_i = k$, and let $U := \text{span}\{f_i : i \in S\} \subseteq \mathbb{F}^{X'}$ (we view each f_i as a function on X' by restriction), so $\dim U \leq |S|$. Its annihilator $W := \{v \in \mathbb{F}^{X'} : \sum_{a \in X'} v(a) f_i(a) = 0 \ \forall i \in S\}$ has $\dim W \geq |X'| - \dim U \geq |X'| - |S|$. A subspace of $\mathbb{F}^{X'}$ of dimension w contains a vector of support size $\geq w$ (in reduced row-echelon form its w pivot columns are distinct coordinates, and the sum of the basis rows is nonzero in each pivot column), so choose $v \in W$ with $|\text{supp } v| \geq \dim W \geq |X'| - |S|$. Contracting D' against v in the last coordinate gives

$$\tilde{D}(x_1, \dots, x_{k-1}) := \sum_{x_k \in X'} v(x_k) D'(x_1, \dots, x_{k-1}, x_k) = \sum_{a \in X'} c_a v(a) \mathbf{1}[x_1 = \dots = x_{k-1} = a],$$

a diagonal $(k - 1)$ -tensor with nonzero diagonal set exactly $\text{supp } v$. On one hand, the $i \in S$ slices vanish under the contraction (their f_i are killed by $v \in W$) while each $i \notin S$ slice contracts to a slice of $\tilde{D}$, so $\text{sr}(\tilde{D}) \leq r - |S|$. On the other hand, the inductive hypothesis gives $\text{sr}(\tilde{D}) \geq |\text{supp } v| \geq |X'| - |S|$. Combining, $|X'| - |S| \leq r - |S|$, i.e. $r \geq |X'|$. As the decomposition was arbitrary, $\text{sr}(D') \geq |X'|$; taking $X' = X$ yields $\text{sr}(D) \geq |X|$, and with the upper bound, $\text{sr}(D) = |X|$.

Part 2.1c (restriction monotonicity). If $T = \sum_{i=1}^r S_i$ with $r = \text{sr}(T)$, then restricting every coordinate to the subsets A_j gives $T|_{\prod_j A_j} = \sum_{i=1}^r (S_i|_{\prod_j A_j})$, and each $S_i|_{\prod_j A_j}$ is again a slice (a restricted product of the form Eq. (1)). Therefore $\text{sr}(T|_{\prod_j A_j}) \leq r = \text{sr}(T)$.

This completes the proof. $\square$

Remark 2.2 (Provenance). The notion of slice rank and the diagonal lower bound 2.1b are due to [4] and are developed systematically in [1]; the lower bound is the symmetric repackaging of the Croot–Lev–Pach rank computation [2]. We reproduce only the two properties 2.1a and 2.1b, together with the restriction monotonicity 2.1c, that the cap-set argument consumes.

3 Upper bound on the slice rank

We now bound the slice rank of the constraint tensor. Define $T : (\mathbb{F}_3^n)^3 \rightarrow \mathbb{F}_3$ by $T(x, y, z) := \mathbf{1}[x + y + z = 0]$, the indicator that the three points sum to zero.

Lemma 3.1 (Upper bound). *The tensor $T(x, y, z) = \mathbf{1}[x + y + z = 0]$ on $(\mathbb{F}_3^n)^3$ admits the polynomial representation*

$$T(x, y, z) = \prod_{i=1}^n (1 - (x_i + y_i + z_i)^2), \quad (2)$$

a reduced polynomial of total degree $2n$, and its slice rank satisfies $\text{sr}(T) \leq 3M_n$, where M_n is as in Definition 1.1.

Proof. The proof has two parts: the polynomial identity Eq. (2), then the slice decomposition obtained by grouping monomials according to their lowest-degree block of variables.

Step 1: the polynomial identity. For $u \in \mathbb{F}_3$, $u^2 = 1$ if $u \neq 0$ and $u^2 = 0$ if $u = 0$, so $1 - u^2 = \mathbf{1}[u = 0]$. Applying this coordinatewise with $u = x_i + y_i + z_i$,

$$\prod_{i=1}^n (1 - (x_i + y_i + z_i)^2) \stackrel{(a)}{=} \prod_{i=1}^n \mathbf{1}[x_i + y_i + z_i = 0] \stackrel{(b)}{=} \mathbf{1}[x + y + z = 0] \stackrel{(c)}{=} T(x, y, z),$$

where (a) is $1 - u^2 = \mathbf{1}[u = 0]$ per coordinate, (b) is that a product of indicators is the indicator of the intersection, and (c) is the definition of T . Each of the n factors has degree 2, so the product is reduced of total degree $2n$.

Step 2: monomial grouping. Expand Eq. (2) into reduced monomials,

$$T(x, y, z) = \sum_{(a,b,c)} \gamma_{a,b,c} x^a y^b z^c, \quad a, b, c \in \{0, 1, 2\}^n, \quad \deg(x^a) + \deg(y^b) + \deg(z^c) \leq 2n, \quad (3)$$

the total-degree bound being inherited from Step 1. Since the three block-degrees of each monomial sum to $\leq 2n$, their minimum is $\leq 2n/3$:

$$\min \left\{ \deg(x^a), \deg(y^b), \deg(z^c) \right\} \leq \frac{2n}{3}. \quad (4)$$

Assign each monomial to the block attaining a degree $\leq 2n/3$ (ties broken by the order $x \prec y \prec z$) and collect class by class:

$$T(x, y, z) = \underbrace{\sum_{\deg(x^a) \leq 2n/3} x^a p_a(y, z)}_{=: T_1} + \underbrace{\sum_{\deg(y^b) \leq 2n/3} y^b q_b(x, z)}_{=: T_2} + \underbrace{\sum_{\deg(z^c) \leq 2n/3} z^c r_c(x, y)}_{=: T_3}, \quad (5)$$

where p_a, q_b, r_c collect the remaining factors; by Eq. (4) each monomial lands in exactly one class, so Eq. (5) is an identity. Each summand $x^a p_a(y, z)$ is a slice in the first coordinate (Eq. (1) with $f = x^a$), and the monomials x^a with $\deg(x^a) \leq 2n/3$ number exactly $M_n = \dim V_{\leq 2n/3}$ (Definition 1.1); thus $\text{sr}(T_1) \leq M_n$, and likewise $\text{sr}(T_2), \text{sr}(T_3) \leq M_n$. By subadditivity Item 2.1a,

$$\text{sr}(T) \leq \text{sr}(T_1) + \text{sr}(T_2) + \text{sr}(T_3) \leq 3M_n,$$

which is the claim. This completes the proof. $\square$

Remark 3.2 (Where the threshold $2n/3$ comes from). The exponent $2n/3$ in Definition 1.1 is exactly $d/3$ for the degree $d = 2n$ of T : with three variable blocks summing to degree $\leq d$, the minimum block-degree is $\leq d/3$, which is the bound Eq. (4). This is the slice-rank (3-tensor) analogue of the Croot–Lev–Pach matrix bound $\text{rank} \leq 2M_{d/2}$ [2], where two variable blocks force a minimum of $\leq d/2$.

4 Lower bound on the slice rank

We use the cap-set hypothesis to show that restricting T to A^3 produces a diagonal tensor of full size, whose slice rank we computed in Item 2.1b.

Lemma 4.1 (Lower bound). *Under Assumption 1.4, the restriction of $T(x, y, z) = \mathbf{1}[x + y + z = 0]$ to A^3 is the diagonal tensor*

$$T|_{A^3}(x, y, z) = \mathbf{1}[x = y = z] = \sum_{a \in A} \mathbf{1}[x = y = z = a], \quad x, y, z \in A, \quad (6)$$

and consequently $\text{sr}(T|_{A^3}) = |A|$.

Proof. We establish the diagonal identity Eq. (6), then read off the slice rank from Item 2.1b.

Step 1: $T|_{A^3}$ is diagonal. Fix $x, y, z \in A$. We claim the constraint collapses to the diagonal:

$$(x + y + z = 0, \quad x, y, z \in A) \iff x = y = z. \quad (7)$$

Here “ $\Leftarrow$ ” holds as $x = y = z = a \Rightarrow x + y + z = 3a = 0$ (since $3 = 0$ in $\mathbb{F}_3$), and “ $\Rightarrow$ ” is precisely Assumption 1.4. Hence

$$T|_{A^3}(x, y, z) \stackrel{(a)}{=} \mathbf{1}[x + y + z = 0] \stackrel{(b)}{=} \mathbf{1}[x = y = z] \stackrel{(c)}{=} \sum_{a \in A} \mathbf{1}[x = y = z = a],$$

where (a) is the definition of T , (b) is Eq. (7), and (c) expands the diagonal indicator over $a \in A$. This is Eq. (6), a diagonal tensor with all coefficients $c_a = 1 \neq 0$.

Step 2: slice rank of the diagonal. Eq. (6) meets the hypotheses of Item 2.1b with $X = A$, $k = 3$, and $c_a = 1 \neq 0$ for all $a \in A$, so

$$\text{sr}(T|_{A^3}) = |A|.$$

This completes the proof. □

Remark 4.2 (The no-progression property is load-bearing). Without Assumption 1.4 the restricted tensor $T|_{A^3}$ would in general have off-diagonal entries (whenever $x + y + z = 0$ with x, y, z not all equal), and the clean identity Eq. (6) would fail; the slice rank could then be far below $|A|$. The cap-set hypothesis is precisely what collapses $T|_{A^3}$ onto its diagonal and forces the lower bound $\text{sr}(T|_{A^3}) \geq |A|$.

5 Counting low-degree monomials

To convert the bound $|A| \leq 3M_n$ into an explicit exponential bound we estimate M_n . The estimate is a one-line Cramér (large-deviation) bound on the coefficients of $(1 + x + x^2)^n$.

Lemma 5.1 (Monomial count). *Let $M_n = \#\{a \in \{0, 1, 2\}^n : \sum_{i=1}^n a_i \leq 2n/3\}$ as in Definition 1.1. Then for every real $t \in (0, 1]$,*

$$M_n \leq t^{-2n/3} (1 + t + t^2)^n, \quad (8)$$

and consequently

$$\limsup_{n \rightarrow \infty} M_n^{1/n} \leq \min_{0 < t \leq 1} \frac{1 + t + t^2}{t^{2/3}} =: 3\gamma, \quad 3\gamma < 2.7558. \quad (9)$$

Proof. Encode the count as the coefficient sum of a generating polynomial, then apply a Markov/Chernoff shift.

Step 1: generating function and Chernoff shift. Since each coordinate a_i ranges over $\{0, 1, 2\}$ independently, the number of $a \in \{0, 1, 2\}^n$ with $\sum_i a_i = s$ equals the coefficient $[u^s](1 + u + u^2)^n$ (the term u^{a_i} contributes from each of the n factors). Hence

$$M_n = \sum_{s=0}^{\lfloor 2n/3 \rfloor} [u^s](1 + u + u^2)^n. \quad (10)$$

Fix $t \in (0, 1]$. For every $s \leq 2n/3$ we have $t^{s-2n/3} \geq 1$, because $s - 2n/3 \leq 0$ and $0 < t \leq 1$. Multiplying each term of Eq. (10) by this factor ≥ 1 and extending the sum to all $s \geq 0$ only increases it:

$$M_n \leq \sum_{s=0}^{\lfloor 2n/3 \rfloor} [u^s](1 + u + u^2)^n \cdot t^{s-2n/3} \leq t^{-2n/3} \sum_{s \geq 0} [u^s](1 + u + u^2)^n t^s = t^{-2n/3} (1 + t + t^2)^n,$$

where the first inequality inserts the factors $t^{s-2n/3} \geq 1$, the second drops the upper limit of summation (all summands are nonnegative since $t > 0$) and pulls out $t^{-2n/3}$, and the last equality evaluates the generating function at $u = t$. This is Eq. (8).

Step 2: optimize the base. Taking n -th roots in Eq. (8) gives, for every $t \in (0, 1]$,

$$M_n^{1/n} \leq \frac{1 + t + t^2}{t^{2/3}}.$$

The right-hand side is independent of n , so taking $\limsup_{n \rightarrow \infty}$ and then the infimum over $t \in (0, 1]$ yields

$$\limsup_{n \rightarrow \infty} M_n^{1/n} \leq \min_{0 < t \leq 1} \frac{1 + t + t^2}{t^{2/3}} =: 3\gamma.$$

The minimand has a unique interior minimizer $t_* \in (0, 1)$, characterized by the first-order condition

$$\frac{d}{dt} \log \frac{1 + t + t^2}{t^{2/3}} = 0 \iff \frac{1 + 2t}{1 + t + t^2} = \frac{2}{3t} \implies t_* \approx 0.5931, \quad 3\gamma = \frac{1 + t_* + t_*^2}{t_*^{2/3}} \approx 2.7551.$$

Evaluating this closed-form optimum, carried out in [3], gives $3\gamma < 2.7558$ (equivalently $\gamma < 0.9183$). This establishes Eq. (9) and completes the proof. $\square$

Remark 5.2 (On citing the numerical optimum). The reduction to the one-variable minimization $\min_{0 < t \leq 1} (1 + t + t^2)t^{-2/3}$ is proved in full above; only the evaluation of the minimizer to the stated decimal places is taken from [3], where the optimization is carried out explicitly. We do not reproduce the floating-point computation, but the value $3\gamma < 2.7558$ is what feeds the corollary.

6 The cap-set bound

Theorem 6.1 (Ellenberg–Gijswijt cap-set bound). *Let $A \subseteq \mathbb{F}_3^n$ be a cap set (Assumption 1.4). Then*

$$|A| \leq 3M_n,$$

where $M_n = \#\{a \in \{0, 1, 2\}^n : \sum_{i=1}^n a_i \leq 2n/3\}$ as in Definition 1.1.

Proof. Chain the two-sided slice-rank bounds on $T(x, y, z) = \mathbf{1}[x + y + z = 0]$, using $A^3 \subseteq (\mathbb{F}_3^n)^3$:

$$|A| \stackrel{(a)}{=} \text{sr}(T|_{A^3}) \stackrel{(b)}{\leq} \text{sr}(T) \stackrel{(c)}{\leq} 3M_n,$$

where (a) is Lemma 4.1, (b) is Item 2.1c, and (c) is Lemma 3.1. This completes the proof. $\square$

Corollary 6.2 (Explicit exponential bound). *There is an absolute constant $C > 0$ such that every cap set $A \subseteq \mathbb{F}_3^n$ satisfies*

$$|A| \leq C \cdot (2.7558)^n,$$

and in particular $\limsup_{n \rightarrow \infty} |A|^{1/n} \leq 3\gamma < 2.7558$, where $3\gamma = \min_{0 < t \leq 1} (1 + t + t^2)t^{-2/3}$ as in Lemma 5.1.

Proof. Combine the combinatorial bound with the monomial count:

$$|A|^{1/n} \stackrel{(a)}{\leq} (3M_n)^{1/n} = 3^{1/n} M_n^{1/n},$$

where (a) is Theorem 6.1. Taking $\limsup_{n \rightarrow \infty}$ and using $3^{1/n} \rightarrow 1$ together with Eq. (9) of Lemma 5.1,

$$\limsup_{n \rightarrow \infty} |A|^{1/n} \leq \left(\lim_{n \rightarrow \infty} 3^{1/n} \right) \cdot \left(\limsup_{n \rightarrow \infty} M_n^{1/n} \right) \leq 1 \cdot 3\gamma = 3\gamma < 2.7558.$$

Since $\limsup_n |A|^{1/n} < 2.7558$, there is an absolute constant $C > 0$ with $|A| \leq C \cdot (2.7558)^n$ for all n (the finitely many n with $|A|^{1/n} \geq 2.7558$, if any, are absorbed into C , and for large n the bound is immediate). This completes the proof. $\square$

Remark 6.3 (Comparison with the literature). Theorem 6.1 is the bound of Ellenberg and Gijswijt [3], obtained from the polynomial method of Croot, Lev, and Pach [2] via the slice-rank formulation of [4, 1]. The base 2.7558 improved on all prior bounds, which were of the form $3^n/n^{1+\epsilon}$, and remains the best known exponential rate for cap sets in $\mathbb{F}_3^n$.

References

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